```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Cleavland(2).csv",header=None)

df.columns = ['age',
        'sex',
        'cp',
        'trestbps',
        'chol',
        'fbs',
        'restecg',
        'thalach',
        'exang',
        'oldpeak',
        'slope',
        'ca',
        'thal',
        'num']

print(df.head())



# a. Histogram
# Objective: To show distribution of patient ages

plt.hist(df['age'])

plt.title("Histogram of Age")

plt.xlabel("Age")

plt.ylabel("Frequency")

plt.show()



# b. Pie Chart
# Objective: To show proportion of heart disease patients
```

```
heart_counts = df['num'].value_counts()

plt.pie(heart_counts,
        labels=heart_counts.index,
        autopct='%1.1f%%')

plt.title("Heart Disease Distribution")

plt.show()


# c. Box Plot
# Objective: To detect spread and outliers in cholesterol

plt.boxplot(df['chol'])

plt.title("Box Plot of Cholesterol")

plt.ylabel("Cholesterol")

plt.show()


# d. Scatter Plot
# Objective: To show relation between age and cholesterol

plt.scatter(df['age'],
            df['chol'])

plt.title("Scatter Plot of Age vs Cholesterol")

plt.xlabel("Age")

plt.ylabel("Cholesterol")

plt.show()


# e. Add Boxplot to Scatterplot
# Objective: To compare data distribution and relationship together
```

```
fig,ax = plt.subplots()

ax.scatter(df['age'],
           df['chol'])

ax.set_title("Scatter Plot with Boxplot")

ax.set_xlabel("Age")

ax.set_ylabel("Cholesterol")

plt.show()

sns.boxplot(x=df['age'])

plt.show()

sns.boxplot(y=df['chol'])

plt.show()
```